

Slide 1 — Title

- NeoBank360: Full Stack Digital Banking Platform
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Slide 2 — Executive Summary

- Modern, scalable digital banking platform for customers + admins
- Four pillars: Account Management, Loan Processing, Financial Analytics, Rewards Management

Slide 3 — 60-Day Roadmap

- High-level sprint timeline from foundation → analytics → deployment

Slide 4 — Technology Stack

- Angular (frontend), Spring Boot (backend APIs), MySQL (database), JWT (secure stateless auth)

Slide 5 — Problem vs. Solution

- Manual onboarding → Digital KYC
- Slow approvals → Automated loan verification
- Limited insights → Real-time analytics
- Fragmented services → One unified platform

Slide 6 — Layered Stack

- Frontend: Angular 21 + Material + Chart.js + TypeScript
- Backend: Spring Boot + Spring Security + JWT + REST
- Data/Docs: MySQL + Swagger/OpenAPI

Slide 7 — System Architecture

- Layered design: Frontend → API → Backend/MySQL
- Independent modules (Auth, Analytics, Notifications, Audit) scale separately

Slide 8 — Core Functional Modules

- Customer Portal: onboarding, transactions, budgets, rewards
- Admin Portal: user management, loan approvals, analytics, compliance

Slide 9 — Authentication & Security

- Zero-trust model: every request authenticated, authorized, audited
- JWT (stateless) + BCrypt (hashing) + RBAC + immutable audit logs

Slide 10 — Account Opening & Banking Ops

- Digital KYC flow: Register → Submit KYC → Upload Docs → Verify → Approve

- Daily banking: transfers, transaction history, balance monitoring, bill payments

Slide 11 — Loan Management Overview

- End-to-end digital lending: products → application → eligibility → approval → tracking → repayment

Slide 12 — Loan Application Workflow

- Visual flow from product selection to repayment, minimizing friction

Slide 13 — Budget & Expense Management

- PFM module: budget creation, auto-categorization, spend tracking, goals, alerts

Slide 14 — Budget Management Workflow

- Step-by-step flow: set goals → track spend → get insights

Slide 15 — Expense Tracking Process

- Transactions → Categorization → Analysis → Budget Impact
- "Spending funnel" shows where money flows

Slide 16 — Financial Insights Generation

- Analytics engine turns raw transactions into actionable insights (compliance-aware)

Slide 17 — Bill Management Module

- Flow: Add Bill → Validate → View Due → Pay
- Ensures no missed payments, full history

Slide 18 — Rewards System

- Auto-awarded points on transactions/activity
- Boosts engagement & customer loyalty

Slide 19 — Financial Calculation Engine

- Core formulas: Budget utilization %, Income/Expense/Savings/Net Worth, EMI (reducing balance), payoff forecasting
- Business value: automated, accurate financial computation

Slide 20 — Admin Management Module

- User management, user control (activate/deactivate/roles), activity monitoring, deposit management
- Drives governance and reliability

Slide 21 — Admin Workflow

- End-to-end onboarding-to-account-management process
- Key benefits: centralized control, efficiency, compliance, real-time monitoring

Slide 22 — Executive Analytics Dashboard

- Transaction analytics (volume, credit/debit trends)
- Loan analytics (approval rate, disbursement, repayment performance)

Slide 23 — Executive Analytics Benefits

- Real-time visibility, risk monitoring, compliance oversight, data-driven decisions, governance

Slide 24 — Executive Analytics Flow

- Raw data → actionable insights pipeline for strategic decisions

Slide 25 — Admin System Audit Logs

- Full operational/request tracking + performance/error diagnostics
- Captures API times, failures, auth issues, root-cause data

Slide 26 — Audit Logs Benefits

- Transparency, faster issue resolution, security/fraud detection, reliability

Slide 27 — Audit Log Flow

- Shows how raw actions become structured, actionable log insights

Slide 28 — Analytics & Monitoring Engine

- Central intelligence hub: transaction, spending, rewards, loan, and executive analytics combined into KPIs

Slide 29 — System Health Monitoring

- Tracks DB status, app status, active sessions, uptime — real-time reliability

Slide 30 — System Health Benefits

- Real-time visibility, fast issue detection, reliability, simplified oversight

Slide 31 — Monitoring Flow

- Visual flow of how health data is captured and surfaced

Slide 32 — Performance, Analytics & Key Learnings

- Challenge: making sense of large data volumes → Solution: multi-type analytics + charts → Learning: built KPI dashboards
- Challenge: data complexity/reliability → Solution: JPA/Hibernate, optimized queries, audit logging → Learning: scalable DB design + observability

Slide 33 — Major Technical Challenges & Solutions

- Security: JWT + Spring Security + route guards → enterprise auth expertise
- Financial logic: EMI engine, budget tracking, net worth calc → applied banking domain knowledge

- API integration: DTOs, standardized responses, Swagger → scalable full-stack architecture

Slide 34 — Closing / Key Achievements

- Recap: auth, onboarding, loans, budgets, bills, rewards, analytics, audit, health monitoring
- Outcome: enterprise-grade full-stack digital banking platform
- Thank You — Questions & Discussion